



QUANTUM MACHINE LEARNING

THE FUTURE OF QML AND PRACTICE

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QUICK CHECK 1: CIRCUIT REASONING



A qubit is initialized in the state $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The following gates are applied in sequence:
 $|0\rangle \rightarrow H \rightarrow Z \rightarrow H$

The gate matrices are $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Which of the following is the final state?

(A) $|0\rangle$

(B) $|1\rangle$

(C) $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

(D) $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

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QUICK CHECK 2: ENCODING OR ANSATZ?



Suppose an encoding generates the Fourier frequency set $\Omega = \{-1, 0, 1\}$. A researcher wants to learn the target function $g(x) = \cos(2x)$. To improve performance, they double the depth of the variational ansatz while keeping the encoding unchanged.

Which statement is correct?

- (A) The deeper ansatz introduces the missing frequency $\omega = 2$, so $g(x)$ can now be represented exactly.
- (B) The encoding already fixes the accessible frequencies. A deeper ansatz can only change the coefficients of the existing frequencies.
- (C) Increasing the ansatz depth always improves generalization, even if the encoding remains unchanged.
- (D) The target function can be represented exactly because $\cos(2x)$ is simply a scaled version of $\cos(x)$.

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QUICK CHECK 3: UNDERSTANDING THE GRAM MATRIX

A QSVM trained on four samples produces the following Gram matrix:

$$K = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Which statement is most accurate?

- (A) The model will generalize well because every training sample is perfectly separated.
- (B) The encoding preserves neighbourhood information, allowing smooth interpolation to unseen samples.
- (C) The encoding behaves like memorization: every sample is only similar to itself, making generalization difficult.
- (D) The kernel is invalid because a Gram matrix cannot be the identity matrix.

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QUICK CHECK 4: SHOT NOISE OR BARREN PLATEAU?

A VQC is trained on a noiseless simulator. As the circuit depth increases from 4 to 20 layers, the gradient variance decreases from 10^{-2} to 10^{-11} . Increasing the number of measurement shots from 1,000 to 100,000 does not improve training.

What is the most likely explanation?

- (A) The optimization failed because the number of shots is too small.
- (B) The circuit is suffering from a barren plateau, where gradients vanish exponentially with depth.
- (C) The kernel has become an identity matrix.
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QUICK CHECK 5: DESIGNING A HYBRID QML MODEL



A team proposes modifying the QXRNet architecture by replacing the linear Q-Mapper with a two-layer nonlinear MLP before the quantum feature map.

Based on the ablation study discussed in class, which statement is most accurate?

- (A) The nonlinear MLP is unnecessary because the quantum feature map already introduces the required nonlinearity.
- (B) The nonlinear MLP should improve performance because more nonlinear layers always increase expressivity.
- (C) The nonlinear MLP reduces the number of trainable parameters, making optimization easier.
- (D) The choice of Q-Mapper has no effect because the PQC dominates the learning process.

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QUICK CHECK 6: INTERPRETING A QUANTUM KERNEL

A QSVM produces the following Gram matrix for four training samples:

$$K = \begin{bmatrix} 1.00 & 0.98 & 0.97 & 0.99 \\ 0.98 & 1.00 & 0.98 & 0.97 \\ 0.97 & 0.98 & 1.00 & 0.98 \\ 0.99 & 0.97 & 0.98 & 1.00 \end{bmatrix}$$

Which conclusion is most appropriate?

- (A) The kernel is excellent because every sample is highly similar to every other sample.
- (B) Increasing the ansatz depth will always improve discrimination.
- (C) The Gram matrix is invalid because all diagonal entries should be zero.
- (D) The feature map has likely suffered from kernel concentration, making different samples difficult to distinguish.

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QUICK CHECK 7: QAOA vs. HARDWARE-EFFICIENT VQCs

Two variational algorithms are designed for the same 20-qubit optimization problem.

Model A: Hardware-efficient VQC, 3 layers, one trainable R_y gate per qubit per layer.

Model B: QAOA with depth $p = 3$.

Which statement is most accurate?

- (A) Both models have the same number of trainable parameters because they use the same circuit depth.
- (B) QAOA has fewer trainable parameters because each layer shares the same γ and β parameters across all qubits.
- (C) The hardware-efficient VQC has fewer trainable parameters because each qubit is optimized independently.
- (D) Both models require the same number of circuit evaluations for one gradient update.

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QUICK CHECK 8: HONEST QUANTUM ADVANTAGE



A paper claims: *“Our quantum algorithm classifies medical images 100× faster than the best classical algorithm.”*

Further reading reveals: the quantum algorithm assumes QRAM; the classical baseline is logistic regression; no comparison is made with quantum-inspired algorithms.

Which statement is the most appropriate conclusion?

- (A) The paper has demonstrated quantum advantage because the reported speedup is large.
- (B) The use of QRAM guarantees exponential quantum advantage.
- (C) The claim is incomplete because the comparison may not be fair without equivalent data-access assumptions and stronger classical baselines.
- (D) The result proves that quantum machine learning is always superior to classical machine learning.

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QUICK CHECK 9: WHAT REALLY CHANGES IN QUANTUM RL?

A classical DQN is converted into a Quantum DQN by replacing the neural network with a Parameterized Quantum Circuit (PQC). The replay buffer, Bellman update rule, optimizer, and environment remain unchanged.

Which statement best explains why this is sufficient?

- (A) The PQC replaces the entire reinforcement learning algorithm, making the replay buffer and Bellman equation unnecessary.
- (B) Reinforcement learning is fundamentally a quantum algorithm; the classical components are retained only for implementation convenience.
- (C) The PQC serves only as the function approximator. The reinforcement learning algorithm itself remains classical and continues to update the PQC parameters.
- (D) Once a PQC is introduced, all computations, including the optimizer and replay buffer, automatically become quantum.

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QUICK CHECK 10: EVALUATING A QML MODEL



Two research groups solve the same classification problem.

Model A

- Accuracy: 98%
- 250 trainable parameters
- 200 qubits
- State preparation: seconds/sample

Model B

- Accuracy: 96%
- 18 trainable parameters
- 8 qubits
- Runs on current NISQ hardware

Which statement is the most appropriate conclusion?

- (A) Model A is unquestionably better because it achieves the highest accuracy.
- (B) Model B is unquestionably better because it uses fewer qubits.
- (C) Model A demonstrates quantum advantage because it uses more qubits.
- (D) Neither model can be declared superior without considering resource requirements, scalability, and the application's constraints.

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THE FUTURE OF QUANTUM MACHINE LEARNING



We now know how to build
Quantum Machine Learning models.

But...

Are today's QML models
good enough?

WHAT WE'VE BUILT – AND WHAT'S STILL OPEN



We can now

- Encode classical data
- Design trainable circuits
- Build hybrid models
- Compute quantum kernels
- Apply QML to RL

Still open

- Scale to large data
- Train deep circuits reliably
- Generalize well
- Work under hardware noise
- Prove real advantage

These open questions define today's research.

Four research directions.

- 1 Can we prove honest quantum advantage?
- 2 Can we design better quantum models?
- 3 Can we measure quantum states efficiently?
- 4 Can QML scale to modern AI?

Let's begin with the most fundamental one.

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FIVE THINGS I HOPE YOU REMEMBER



- ① Think before choosing an encoding – it fixes what your model can learn
- ② Bigger circuits are not always better – structure often beats brute force
- ③ Quantum advantage needs a fair, same-access-model comparison
- ④ Trainability and expressivity are two sides of the same coin
- ⑤ Today's limitations are tomorrow's research opportunities

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Q1: QUANTUM STATE PREPARATION AND SIMILARITY

Scenario. A quantum sensing system monitors two chemical samples, encoded into a 1-qubit feature map using angle encoding.

Sample A: $x = \left(0, \frac{\pi}{3}\right)$ Sample B: $x' = \left(0, \frac{2\pi}{3}\right)$

Before encoding, the calibration circuit prepares $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$.

- (a) Verify whether $|\psi\rangle$ is a valid quantum state by checking normalization. [2 Marks]
- (b) For the two encoded samples, compute the kernel $K(x, x') = \cos^2\left(\frac{x - x'}{2}\right)$ and interpret the result. [2 Marks]
- (c) If basis encoding were used instead of angle encoding, explain how the kernel value and the model's ability to generalize would change. [2 Marks]

Q2: DESIGNING A TRAINABLE HYBRID MODEL



Scenario. A hospital develops a hybrid classifier: 48-dimensional features $\rightarrow$ Q-Mapper

The Q-Mapper is linear. The VQC contains 3 layers, with one R_y rotation per qubit per layer.

- (a) Compute (i) Q-Mapper parameters, (ii) VQC parameters, (iii) classical head parameters, (iv) total parameters. [2 Marks]
- (b) The team proposes replacing the linear Q-Mapper with a two-layer nonlinear MLP before quantum encoding. Should they do this? Explain. [2 Marks]
- (c) After increasing the circuit depth from 3 to 18 layers, training almost stops although the optimizer is unchanged. Identify the likely reason and suggest one improvement. [2 Marks]

Q3: EVALUATING A QUANTUM AI CLAIM



Scenario. A logistics company reports: “Our Quantum Reinforcement Learning system trains warehouse robots $80\times$ faster than the best classical method.” Investigation reveals the policy network is replaced by a PQC, the replay buffer and Bellman update remain unchanged, and QRAM is assumed.

- (a) Which parts of the DQN pipeline become quantum, and which remain classical? [2 Marks]
- (b) Before accepting the claimed quantum advantage, list three aspects that should be verified. [2 Marks]
- (c) Suppose another research group develops a classical algorithm using the same data-access assumptions and achieves similar runtime. What conclusion should be drawn? [2 Marks]



Thank you :)